Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering and Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: CNN**	
Experiment No:- 4	**Date:-**	**Enrollment No:-**92301733001

Code:

Exp4.ipynb

```
model = models.Sequential([
    layers.Conv2D(32, (3, 3), activation='relu', input_shape=(32, 32, 3)),
    layers.MaxPooling2D(pool_size=(2, 2)),
    layers.Flatten(),
    layers.Dense(128, activation='relu'),
    layers.Dense(10, activation='softmax')
])

model.compile(optimizer='adam',
              loss='categorical_crossentropy',
              metrics=['accuracy'])

history = model.fit(train_images, train_labels, epochs=10, batch_size=64, validation_data=(test_images, test_labels))

test_loss, test_acc = model.evaluate(test_images, test_labels)
print(f"Test accuracy: {test_acc * 100:.2f}%")

plt.plot(history.history['accuracy'], label='Training Accuracy')
plt.plot(history.history['val_accuracy'], label='Validation Accuracy')
plt.title('Model Accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.legend(loc='lower right')
plt.show()

plt.plot(history.history['loss'], label='Training Loss')
plt.plot(history.history['val_loss'], label='Validation Loss')
plt.title('Model Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.legend(loc='upper right')
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```
plt.show()
```

```
Downloading data from https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz
170498071/170498071 ━━━━━━━━━━━━━━━━━━━━ 12s 0us/step
/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarning: Do not pass an `input_shape`/`input_dim` arg
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Epoch 1/10
782/782 ━━━━━━━━━━━━━━━━━━━━ 42s 51ms/step - accuracy: 0.4650 - loss: 1.5011 - val_accuracy: 0.5378 - val_loss: 1.3101
Epoch 2/10
782/782 ━━━━━━━━━━━━━━━━━━━━ 40s 51ms/step - accuracy: 0.5757 - loss: 1.2062 - val_accuracy: 0.5615 - val_loss: 1.2606
Epoch 3/10
782/782 ━━━━━━━━━━━━━━━━━━━━ 40s 51ms/step - accuracy: 0.6159 - loss: 1.0991 - val_accuracy: 0.6163 - val_loss: 1.1061
Epoch 4/10
782/782 ━━━━━━━━━━━━━━━━━━━━ 41s 51ms/step - accuracy: 0.6475 - loss: 1.0138 - val_accuracy: 0.6244 - val_loss: 1.0695
Epoch 5/10
782/782 ━━━━━━━━━━━━━━━━━━━━ 42s 52ms/step - accuracy: 0.6730 - loss: 0.9377 - val_accuracy: 0.6357 - val_loss: 1.0473
Epoch 6/10
782/782 ━━━━━━━━━━━━━━━━━━━━ 40s 51ms/step - accuracy: 0.6948 - loss: 0.8768 - val_accuracy: 0.6398 - val_loss: 1.0301
Epoch 7/10
782/782 ━━━━━━━━━━━━━━━━━━━━ 40s 52ms/step - accuracy: 0.7147 - loss: 0.8235 - val_accuracy: 0.6524 - val_loss: 0.9978
Epoch 8/10
782/782 ━━━━━━━━━━━━━━━━━━━━ 40s 51ms/step - accuracy: 0.7296 - loss: 0.7744 - val_accuracy: 0.6566 - val_loss: 1.0051
Epoch 9/10
782/782 ━━━━━━━━━━━━━━━━━━━━ 41s 51ms/step - accuracy: 0.7471 - loss: 0.7280 - val_accuracy: 0.6447 - val_loss: 1.0859
Epoch 10/10
782/782 ━━━━━━━━━━━━━━━━━━━━ 41s 52ms/step - accuracy: 0.7612 - loss: 0.6850 - val_accuracy: 0.6441 - val_loss: 1.0715
313/313 ━━━━━━━━━━━━━━━━━━━━ 3s 9ms/step - accuracy: 0.6441 - loss: 1.0715
Test accuracy: 64.41%
```

Exp4.ipynb

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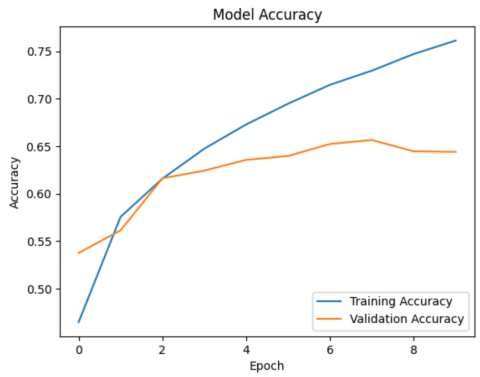
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